

Contact

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Portfolio: walid-birouk.github.io/portfolio

Expertise & Skills

AI / ML
- Generative AI, RAG, LLM agents, TensorFlow, YOLOv8, U-Net, autoencoders

Languages & data
- Python, SQL, PostgreSQL, Supabase, BigQuery, dbt, Azure Databricks

Pipelines
- Spark, Spark Structured Streaming, Kafka, Delta Lake, Airflow

Cloud & MLOps
- Google Cloud (Vertex AI), AWS (SageMaker), Azure, Dataiku, Git

Other
- HubSpot integration, requirements analysis, data modelling, production support

Languages

English: Fluent
French: Professional working proficiency
Dutch: Basic (A2)
Amazigh: Native

Certifications

Dataiku
Generative AI Practitioner, ML Practitioner, MLOps Practitioner, Core Designer, Advanced Designer, Developer

AWS Academy
Data Engineering, Graduate (40 h)

Microsoft / Coursera
Azure Databricks for Data Engineering

IBM / Coursera
Machine Learning with Python

dbt Labs
dbt Fundamentals

WALID BIROUK

AI & DATA ENGINEER

About Me

AI & Data Engineer with hands-on experience in **generative AI** (RAG, LLM agents) and **data engineering** on Google Cloud, AWS and PostgreSQL. I have built AI assistants integrated into business tools, ML systems for railway safety, and the data model and automations of a production e-commerce store. I work freelance on AI automation and am open to a full-time role as an AI, ML or Data Engineer.

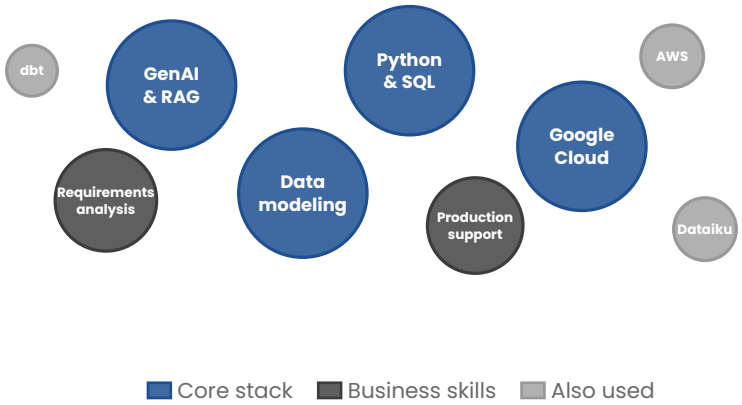
Experience

Data Engineer, Ghazeline 2026/01 – Present
Designed the PostgreSQL / Supabase data model (products, variants, categories, stock, orders) for the online store of a ready-to-wear fashion brand. Built backend features for data collection and automated the creation of order summary sheets, and deployed an n8n AI agent that automates Instagram replies. I maintain the store and ship fixes and new features from business feedback.

ML Engineer, RideOnTrack (Infrabel project) 2025/01 – 2025/08
Developed an anomaly detection system for Belgian railway tracks with machine learning, using a U-Net segmentation model and an autoencoder to isolate critical areas. Integrated YOLOv8 for real-time train signal recognition (91% accuracy).

Generative AI Engineer, Algorhythm 2024/09 – 2024/12
Developed AI workflows on Vertex AI and BigQuery to automate internal tasks with a consistent, specialised tone. Built a RAG agent over internal BigQuery data and integrated it into HubSpot to support internal communication and automate repetitive queries.

Data Engineering & Data Quality Intern, Algorhythm 2024/02 – 2024/07
Built a data observability proof of concept: automated dbt tests and manual data-quality checks, with results piped to OpenMetadata.



Selected Projects

Ghazeline Shop 2026/01 – Present
Ghazeline (ready-to-wear brand) · Role: Data Engineer (freelance) · Industry: Fashion retail
- Designed the PostgreSQL / Supabase data model: products, variants, categories, stock and orders
- Built backend features that collect data into that model
- Automated the creation of order summary sheets
- Deployed an **n8n AI agent** that replies to Instagram messages
- Maintain the store: fixes and new features from business feedback

Tech stack: PostgreSQL, Supabase, n8n

RAG Agent in HubSpot 2024/09 – 2024/12
Algorhythm (internal) · Role: Generative AI Engineer
- Built a **RAG agent** that answers from internal BigQuery data
- Integrated it into **HubSpot** to automate repetitive internal queries
- Developed Vertex AI workflows that keep a consistent, specialised tone

Tech stack: Vertex AI, BigQuery, HubSpot

Railway Track Anomaly Detection 2025/01 – 2025/08
Infrabel (via RideOnTrack) · Role: ML Engineer · Industry: Railways
- Segmented track images with **U-Net** and an **autoencoder** to isolate critical areas
- Real-time train signal recognition with **YOLOv8**, **91% accuracy**

Tech stack: Python, TensorFlow, U-Net, YOLOv8, AWS SageMaker, Git

All projects on my website: walid-birouk.github.io/portfolio

Education

Master's programme, Information Systems & AI 2025 – 2026
EPSI, France

Bachelor's, Data & Artificial Intelligence 2021 – 2024
Thomas More, Mechelen, Belgium

Licence, Mathematics & Computer Science 2019 – 2021
Université de Jijel, Algeria